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Optimizing AI Industry

To optimize the intelligence of AI actions, neural networks must act according to the principles of epistemology.

Therefore, neural networks must receive learning and analyzed text information in the format of semantic units – descriptions of elementary properties of objects, operate with these units, and generate output messages in the format of semantic units.

Using information formatted according to the principles of epistemology will significantly increase the profitability of the production and application of AI.

One of the main functions of AI should be the detection of semantic errors in analyzed texts – incorrect words, logical contradictions, and inconsistencies with facts.

To minimize the likelihood of generating errors and hallucinations, AI must function in a discussion mode.

AI bot messages should be checked by another AI bot or even two bots to detect errors.

AI bots should present discussions of their errors to users, who will participate in the discussions and determine the persuasiveness of the bots' arguments.

AI uses both reliable information and false information and often does not distinguish true information from false.

If several alternative hypotheses on the content of the output message are known, the AI should communicate all such hypotheses to users, who will determine the persuasiveness and possible erroneousness of the hypotheses.

AI should collect user assessments of the credibility of information sources and communicate them to users.

For the effectiveness of AI-user interactions to be optimal, users must adhere to the principles of epistemology.

Therefore, the dissemination of AI technologies should be accompanied by the propagation of scientific formatting of semantic information and the formalization of the intellectual activity of users.

Manufacturers of AI technologies should include descriptions of these procedures in the manuals for using these technologies.

Corporations producing AI systems must persuade the US corporate business community to implement these epistemic procedures by promoting their intelligent process practices.

To do this, they must implement these procedures in their internal activities – in document formats, in management processes, and in decision making – and publish information about these procedures and their high efficiency.

The development of AI technologies will stimulate the application of the **principles of epistemology in all intellectual fields** – in economics, science, education, etc.

Mass compliance with these principles will cause an economic effect greater than the use of AI technologies.

The joint use of the principles of epistemology and AI technologies will create their additional synergistic effect.

The spread of AI will qualitatively change not only technologies and businesses but even social systems.

Therefore, the US Government should create the **Public Council to optimize the social effects of AI.**

Members of this Council should be corporations – creators of AI technologies, scientific institutes, representatives of the Government, business associations, professional unions.

This Council will create bills and recommendations for the development

and investment of AI technologies.

This Council will operate on the Internet, constantly, based on the principles of epistemology.

The possible unprofitability of AI technologies will cause not only losses to investors but also the impossibility of recouping part of the AI industry.

Therefore, part of the AI data centers will be non-profit and public.

They will belong to social associations, municipal organizations, and government agencies.

One of the main intellectual functions of people and social systems is the **Choice**.

People, businesses, and societies spend a lot of effort, time, and resources on this function.

People choose goods and services, places of work and residence, marriage partners, doctors, lawyers.

Businesses choose partners, business models, products, technologies, locations, logistics, workers.

Societies choose places to build new settlements, objects in them, production, transport routes, infrastructure.

AI neural networks are able to effectively search for options for ways to satisfy these needs, detect possible errors in them and reject erroneous options, compare options according to utility criteria and recommend the

best ones.

These functions are relatively simpler and more formalizable compared to complex planning and innovation management processes.

At the same time, these functions are very widespread and can generate profits.

Thus, the Choice function should be the primary and main direction of development of commercial AI technologies.

The archaic US PTO patent system causes \$ hundreds of billions in losses to the US economy every year.

AI will not be able to operate effectively in the existing system due to its cumbersomeness and overflow of low-quality patent information.

Reforming the patent system according to the principles of epistemology will maximize the effectiveness of AI in this system and significantly improve the functioning of the technological sectors of the economy.

"The Scientific System of Collective Discussions and Evaluations"

will provide opportunities to adequately assess the reliability, value, and errors of scientific papers and information sources and will increase the effectiveness of scientific activity several times.

The scientific community sabotages the creation of such systems

because they will clearly show the triviality, insignificance, and erroneousness of most scientific papers.

Corporations that create AI technologies are interested in using reliable knowledge and minimizing false knowledge to learn neural networks; therefore, they should initiate the creation and widespread use of such systems.

In order to minimize layoffs, professional associations of specialists sabotage the use of AI and falsify sham errors in its work.

Doctors' associations are the main obstacle to the use of AI in medicine. They claim that AI is not capable of diagnosing diseases and recommending treatments. Although today, AI determines correct diagnoses and treatments twice as well as most doctors.

The main method to overcome such sabotage is the creation of consumer and business associations that will hire competent and objective experts and, according to their conclusions, will license AI technologies and systems, including medical ones.

Higher schools, colleges, and universities will stop archaic and ineffective lecture teaching.

They will mainly train students' practical skills.

AI tutors will personally give theoretical knowledge to students.

Students will practice in production, in offices, in research institutes.

Such education will maximize efficiency and minimize cost.

Scientific functions will be separated from universities and transferred to scientific institutes.

Multimodal AI polygraphs will simultaneously analyze physiological processes, facial expressions, voice, handwriting, gestures, motor skills, and brain MRI of examinees.

Such AI polygraphs will reliably detect lies with a probability of 99.9%. Their widespread use will minimize crime, fraud, and espionage to almost zero.

Respectfully, E. Gershman
